

Github Link: <https://github.com/amytan830/Data-200-Lab-2--Stock-Analysis>

stock_GUI.py

Description	Output	
Menu Screen Added stocks	(myname) Stock Manager Microsoft - 10.0 Shares MSFT GOOGL AMZN Main History Report Symbol Name Shares Add Stock Delete Stock Update Shares Buy Shares Sell Shares Scrape data from Yahoo Finance from Web menu (01/01/25 -> 01/31/25)	Microsoft - 10.0 Shares MSFT GOOGL AMZN Main History Report - Date - - Price - - Volume - ===== 01/02/25 \$414.57 16896500.0 01/03/25 \$419.29 16662900.0 01/06/25 \$423.75 20573600.0 01/07/25 \$418.32 18139100.0 01/08/25 \$420.49 15054600.0 01/10/25 \$414.94 20201100.0 01/13/25 \$413.19 17604800.0 01/14/25 \$411.69 16935900.0 01/15/25 \$422.22 19637800.0 01/16/25 \$420.51 15300000.0 01/17/25 \$424.92 26197500.0 01/21/25 \$424.39 26085700.0 01/22/25 \$441.92 27803800.0 01/23/25 \$442.43 18389300.0 01/24/25 \$439.80 15549500.0 01/27/25 \$430.40 35647800.0 01/28/25 \$442.91 23491700.0 01/29/25 \$438.09 23581400.0 01/30/25 \$411.01 54586300.0

Report

Microsoft - 10.0 Shares

MSFT

GOOGL

AMZN

TSLA

AAPL

WMT

MainHistoryReport

Profit/Loss Report

=====

Symbol: MSFT

Shares Owned: 10.0

First Price: \$414.57

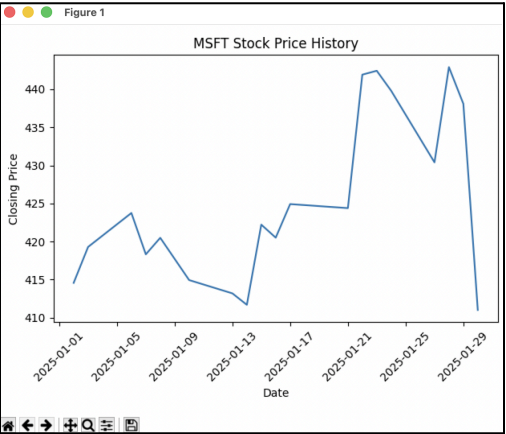
Latest Price: \$411.01

Change Per Share: \$-3.56

Total Gain/Loss: \$-35.60

Percent Change: -0.86%

Created a chart for MSFT



2.3 Test GUI CSV Import Code
WMT stock was added (no existing data)

Walmart Inc. - 26.0 Shares

MSFT

GOOGL

AMZN

WMT

MainHistoryReport

- Date - - Price - - Volume -

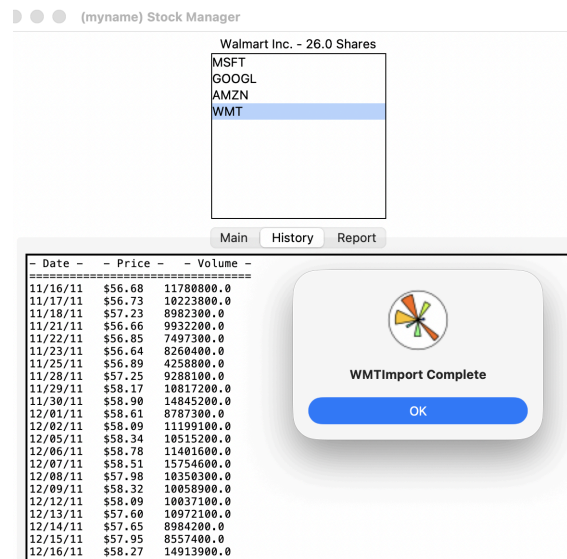
=====

Import from CSV

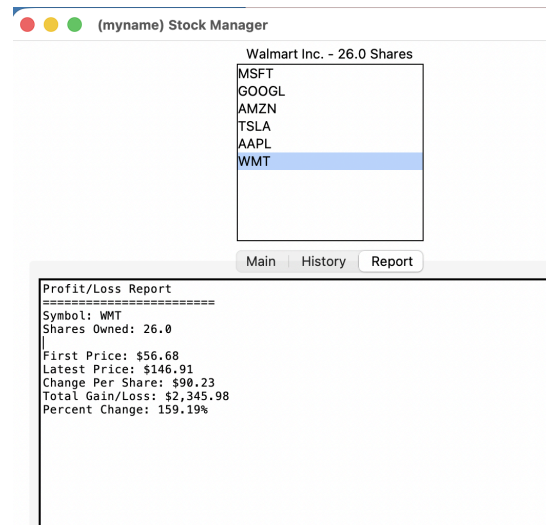
Since Yahoo! Finance does not allow us to download historical data without paying, I found a dataset on Kaggle that contains Walmart stock data from 2011- 2021 and was obtained from Yahoo! Finance.

Link to dataset:

<https://www.kaggle.com/datasets/namithadeshpande027/walmart-stock-historical-data>

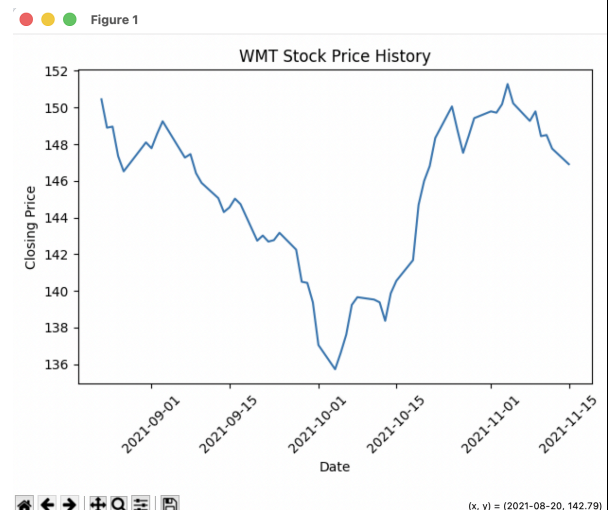


Check Report for WMT after CSV import



Create chart for imported data from CSV

- Since there are 10 years of data, I set display_stock_chart() to only show the latest 60 days of data in the chart



stock_console.py

Description	Output
Main Screen	<pre>amytan@amys-mbp / % cd "/Users/amytan/Desktop/Data 200 - Python/Lab 2" python3 stock_console.py Stock Analyzer --- 1 - Manage Stocks (Add, Update, Delete, List) 2 - Add Daily Stock Data (Date, Price, Volume) 3 - Show Report 4 - Show Chart 5 - Manage Data (Save, Load, Retrieve, Import) 0 - Exit Program Enter Menu Option: █</pre>
Menu Option 1 - Manage Stocks -> Option 4 List Stocks to see current stocks list	<pre>Manage Stocks --- 1 - Add Stock 2 - Update Shares 3 - Delete Stock 4 - List Stocks 0 - Exit Manage Stocks Enter Menu Option: 4 List Stocks --- MSFT (Microsoft) - Shares: 10.0 GOOGL (Google) - Shares: 20.0 AMZN (Amazon) - Shares: 20.0 WMT (Walmart Inc.) - Shares: 26.0 Press Enter to Continue█</pre>
Add Stock	<pre>Manage Stocks --- 1 - Add Stock 2 - Update Shares 3 - Delete Stock 4 - List Stocks 0 - Exit Manage Stocks Enter Menu Option: 1 Add Stock --- Enter stock symbol: AAPL Enter company name: Apple Enter shares: 123 AAPL added. Press Enter to continue█</pre>
Update Shares - Buy Shares	<pre>Update Shares --- 1 - Buy Shares 2 - Sell Shares 0 - Return to Main Menu Enter Menu Option: 1 Buy Shares --- Stock List: [MSFT GOOGL AMZN WMT AAPL] Enter Stock Symbol: AAPL Enter Shares to Buy: 20 Shares purchased.█</pre>

	<pre> Update Shares ---- 1 - Buy Shares 2 - Sell Shares 0 - Return to Main Menu Enter Menu Option: █ </pre>
Update Shares - Sell shares	<pre> Update Shares --- 1 - Buy Shares 2 - Sell Shares 0 - Return to Main Menu Enter Menu Option: 2 MSFT GOOGL AMZN WMT AAPL] Enter Stock Symbol: AMZN Enter Shares to Sell: 1 Shares sold. </pre>
Delete stock	<pre> Manage Stocks --- 1 - Add Stock 2 - Update Shares 3 - Delete Stock 4 - List Stocks 0 - Exit Manage Stocks Enter Menu Option: 3 Delete Stock ---- Stock List: [MSFT GOOGL AMZN WMT AAPL] Enter Stock Symbol to Delete: GOOGL GOOGL deleted. </pre>
List Stocks - GOOGL no longer in our stock list because we deleted - AAPL has 143 shares because we added 20 shares - AMZN has 19 shares because we sold 1 share	<pre> Manage Stocks ---- 1 - Add Stock 2 - Update Shares 3 - Delete Stock 4 - List Stocks 0 - Exit Manage Stocks Enter Menu Option: 4 List Stocks --- MSFT (Microsoft) - Shares: 10.0 AMZN (Amazon) - Shares: 19.0 WMT (Walmart Inc.) - Shares: 26.0 AAPL (Apple) - Shares: 143.0 Press Enter to Continue█ </pre>

Menu option 2 - Add Daily Stock Data

```
2 - Add Daily Stock Data (Date, Price, Volume)
3 - Show Report
4 - Show Chart
5 - Manage Data (Save, Load, Retrieve, Import)
0 - Exit Program
Enter Menu Option: 2
Add Daily Stock Data ---
Stock List: [MSFT AMZN WMT AAPL ]
Enter Stock Symbol: MSFT
Enter Date (m/d/yy): 2/01/25
Enter Closing Price: 412.00
Enter Volume: 2589098
Daily data added.
Press Enter to Continue
```

Menu 3 - Show Report

- We can see that Microsoft's latest price is now \$412.00 since the daily stock data for 2/1/25 we added in menu option 2 above is the latest date (previous web scraping only had dating from 1/1/25 to 1/31/25)

```
Enter Menu Option: 3
Stock Report ---

MSFT - Microsoft
Shares: 10.0
First Price: $414.57
Latest Price: $412.00
Gain/Loss: $-25.70

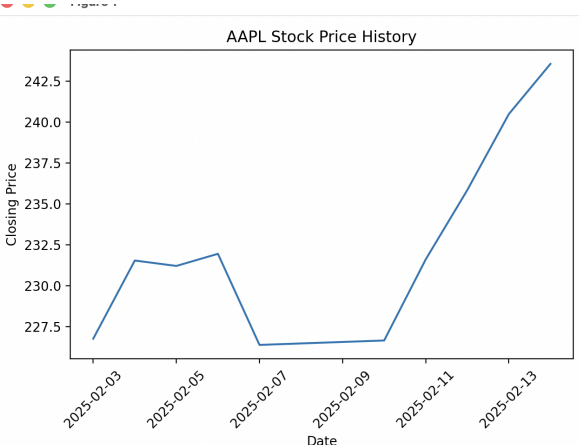
AMZN - Amazon
Shares: 19.0
First Price: $220.22
Latest Price: $234.64
Gain/Loss: $273.98

WMT - Walmart Inc.
Shares: 26.0
First Price: $56.68
Latest Price: $146.91
Gain/Loss: $2,345.98

AAPL - Apple
Shares: 143.0
No daily data available.
Press Enter to Continue
```

Menu Option 5- Manage Data retrieve_from_web()

```
Enter Menu Option: 3
Retrieving Stock Data from Yahoo! Finance ---
This will retrieve data from all stocks in your stock list.
Enter Starting Date (m/d/yy): 2/2/25
Enter Ending Date (m/d/yy): 2/15/25
Records Retrieved: 40
```

Save and Load Data	<pre> Manage Data --- 1 - Save Data 2 - Load Data 3 - Retrieve Data from Web 4 - Import from CSV File 0 - Exit Manage Data Enter Menu Option: 1 Data Saved. </pre> <pre> Manage Data --- 1 - Save Data 2 - Load Data 3 - Retrieve Data from Web 4 - Import from CSV File 0 - Exit Manage Data Enter Menu Option: 2 Data Loaded. </pre>																								
Show Report after Scrapping data - We can now see that there is data available for AAPL	<pre> AAPL - Apple Shares: 143.0 First Price: \$226.77 Latest Price: \$243.54 Gain/Loss: \$2,398.11 Press Enter to Continue </pre>																								
Show Chart after web scraping - Since APPL was added after all the other stocks, it only has data from the web scraping from 2/2/25 to 2/15/25	 <table border="1"> <caption>AAPL Stock Price History Data</caption> <thead> <tr> <th>Date</th> <th>Closing Price</th> </tr> </thead> <tbody> <tr><td>2025-02-03</td><td>227.5</td></tr> <tr><td>2025-02-04</td><td>232.5</td></tr> <tr><td>2025-02-05</td><td>231.5</td></tr> <tr><td>2025-02-06</td><td>232.5</td></tr> <tr><td>2025-02-07</td><td>227.5</td></tr> <tr><td>2025-02-08</td><td>227.5</td></tr> <tr><td>2025-02-09</td><td>227.5</td></tr> <tr><td>2025-02-10</td><td>227.5</td></tr> <tr><td>2025-02-11</td><td>232.5</td></tr> <tr><td>2025-02-12</td><td>240.0</td></tr> <tr><td>2025-02-13</td><td>242.5</td></tr> </tbody> </table>	Date	Closing Price	2025-02-03	227.5	2025-02-04	232.5	2025-02-05	231.5	2025-02-06	232.5	2025-02-07	227.5	2025-02-08	227.5	2025-02-09	227.5	2025-02-10	227.5	2025-02-11	232.5	2025-02-12	240.0	2025-02-13	242.5
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Import CSV stock_console.py test

1. Add new stock

```
Add Stock ---
Enter stock symbol: TSLA
Enter company name: Tesla, Inc.
Enter shares: 20
TSLA added.
Press Enter to continue
```

```
TSLA - Tesla, Inc.
Shares: 20.0
No daily data available.
Press Enter to Continue
```

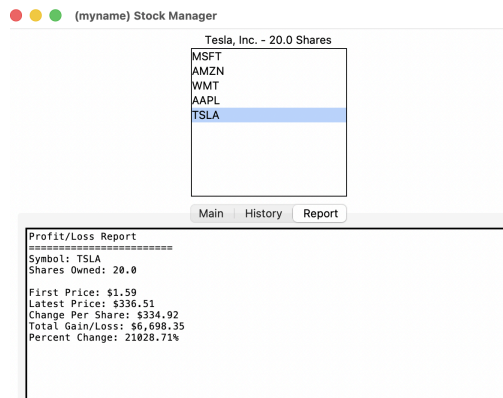
2. Import data for the new stock from csv file

I imported a TSLA dataset from Kaggle, which follows the same format as Yahoo Finance stock data: <https://www.kaggle.com/datasets/umerhaddii/tesla-stock-data-2025>

- The CSV has 3,680 rows with data from 2010-06-29 to 2025-02-12

```
Enter Menu Option: 4
Import CSV from Yahoo! Finance ---
Stock List: [MSFT AMZN WMT AAPL TSLA ]
Enter Stock Symbol: TSLA
Enter CSV filename with full path: /Users/amytan/Desktop/Data 200 - Python/Lab 2/TSLA.csv
CSV Imported. Records Imported: 3680
Press Enter to Continue
```

Confirm that TSLA now has data in the report from the CSV import.



```
TSLA - Tesla, Inc.
Shares: 20.0
First Price: $1.59
Latest Price: $336.51
Gain/Loss: $6,698.35
Press Enter to Continue
```